



OPEN Spectrally optimised YOLOv10s-SeqOpt framework for real-time UAV-based early detection of avocado foliar diseases in indian orchards

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Avocado orchards in southern India experience annual yield losses of 20–40% due to foliar pathogens, including Anthracnose (*Colletotrichum Gloeosporioides*), Root Rot (*Phytophthora Cinnamomi*), Algal Leaf Spot (*Cephaleuros Virescens*) and Scab (*Elsinoe Perseae*). Conventional field scouting and RGB based imaging are unable to detect early-stage infections beneath dense canopy cover, delaying timely intervention and contributing to excessive fungicide use. To address these limitations, this study introduces a UAV based multispectral-RGB imaging pipeline paired with a deep learning detection framework built from the YOLO family of models (YOLOv5s-YOLOv10s) evaluated on a field-collected dataset of 5,953 annotated images comprising 35,422 bounding box instances across four foliar disease classes. A Sequential SGD-AdamW (SeqOpt) optimizer was designed to stabilize gradient dynamics multispectral noise by combining the global exploration capability of SGD in early training phases with the adaptive refinement precision of AdamW in later phases. As a result, SeqOpt achieved an 8% reduction in final loss and a 14% improvement in stability index compared to individual baseline optimisers, confirming superior convergence quality under spectrally heterogeneous training conditions. Systematic benchmarking across all six architectures demonstrated that YOLOv10s-SeqOpt delivered the highest detection performance, achieving 96.0% accuracy, F1-score = 0.911, and mAP@0.5 = 0.937 on multispectral validation set. YOLOv10s-SeqOpt consistently outperformed YOLOv5s-YOLOv9s across both RGB and multispectral datasets, with multispectral OCN input providing measurable detection advantages over RGB across five of six architectures evaluated. The optimized model was quantized and deployed on two embedded AI platforms - NVIDIA Jetson Orin Nano, achieving real-time inference at 69.5ms per frame under PyTorch runtime, and the Raspberry Pi 5, supporting scheduled survey inference at 1.545s per frame via ONNX runtime - both without thermal throttling across extended evaluation sessions. The proposed framework demonstrates field-ready capability for large-scale orchard monitoring, bridging the gap between research-grade detection models and real-world precision agriculture deployment through a validated, scalable pipeline integrating multispectral imaging, deep learning and embedded edge computing for sustainable disease management in Indian horticulture.

Keywords Avocado leaf diseases, UAV imaging, Multispectral analysis, YOLOv10, Sequential SGD-AdamW optimiser (SeqOpt), Edge AI deployment, Precision agriculture, Real-time disease detection.

Avocado (*Persea americana* Mill.) farming is expanding rapidly across the southern states of India, as demand increases for high-value horticultural products. However, its productivity is threatened by several major foliar diseases, including *Colletotrichum gloeosporioides* (Anthracnose), *Phytophthora cinnamomi* (Root Rot), *Cephaleuros virescens* (Algal Leaf Spot), and *Elsinoe perseae* (Scab). These pathogens are well-adapted to the humid subtropical climate of the region and account for annual yield losses of an estimated 20–40%, which

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is nearly ₹150–180 crore in associated economic losses. Conventional disease management relies on manual field scouting and RGB-based image surveys^{1–3} both of which are labour-intensive and incapable of detecting infections before canopy-level symptoms become clearly visible. Consequently, early disease identification for timely fungicide application and canopy management continues to be a bottleneck in avocado production systems across India.

Remote sensing technologies and artificial intelligence (AI) offer considerable potential for precision agriculture applications, including to monitor stress in crops, nutrient deficiency, and disease emergence. Within AI-based approaches, convolutional neural networks (CNNs) and transformer architectures have demonstrated strong performance in plant pathology^{2,4–6}. However, the large model size, excessive power consumption, and dependence on cloud resources, limit their deployment in UAV-based real-time situations. Furthermore, dense avocado canopies present spectral and geometric challenges - occlusions, illumination variability, and poor lesions to background spectral contrast - that further reduce the effectiveness of standard RGB models. The YOLO (You Only Look Once) family of one-stage object detectors has gained widespread adoption for agricultural vision tasks due to ability to perform simultaneous object localisation and classification with low inference latency.

The evolution of YOLO from version 5 to version 10^{7–13} reflects substantial architectural improvements involving feature aggregation, transformer-assisted contextual reasoning, and decoupling of detection heads, alongside improved suitability for deployment in-field^{14–17}. Nevertheless, while prior work has addressed disease detection in apple, grape, maize and tomato crops, no study has yet conducted a comprehensive benchmark of YOLOv5-YOLOv10 architectures under multispectral UAV imagery of avocado foliar disease detection. Additionally, standard optimisation algorithms such as SGD and AdamW tend to exhibit instability under multispectral noise due to gradient oscillation between spectrally dissimilar channels, leading to poor convergence and unstable learning of lesion boundaries.

To overcome these limitations, this study introduces a spectrally optimised YOLOv10s-SeqOpt framework for the early detection of avocado foliar disease from UAV imagery. Specifically, this study proposes a Sequential SGD-AdamW (SeqOpt) optimiser designed to stabilise gradient dynamics by combining the global exploration properties of SGD with fine-grained adaptive refinement of AdamW. This dual-phase learning improves training stability and validation loss during convergence, especially on spectrally heterogeneous multispectral datasets. The framework integrates UAV-acquired multispectral OCN and RGB imagery and evaluated on field collected dataset of 5,953 annotated image tiles comprising 35,422 bounding box instances across four foliar disease classes. The framework is validated for real-time edge deployment on the NVIDIA Jetson Orin Nano (Jetson) and Raspberry Pi 5 (RPi5), providing low-power alternative to cloud inference infrastructure for field-level disease monitoring. The resulting detection pipeline supports proactive management decisions, enabling targeted fungicide applications, and contributing to resource sustainability in Indian orchards. By bridging the gap between fundamental deep learning models developed under laboratory conditions and deployable field-based systems, this study presents a novel, scalable, and cost-effective framework for precision horticulture.

Related work and literature gap

Recent progress in deep learning and remote sensing has substantially advanced crop health monitoring capabilities, with growing research attention directed toward high-value horticultural crops such as avocado (*Persea americana*). Multiple studies have reported advances in multispectral imaging, UAV based remote sensing and lightweight neural networks architecture for agricultural vision applications over the past two years. However, no published agricultural vision study has addressed an end-to-end pipeline integrating UAV multispectral imaging, YOLO-based detection architectures, domain specific optimiser design, and edge deployment specifically for avocado foliar diseases detection. Duncan et al. (2024)¹⁸ proposed a framework aimed at the spectral assessment of *Phytophthora* root rot severity in avocado orchard, employing time-series remote sensing indices to map canopy decline at plantation scale. Although this method was effective in showing the potential for spectral data in large scale canopy health assessment, it did not include object level lesion detection and deep learning models. Hence, this method is not suitable for early-stage fine grained disease identification at the lesion level.

Chávez-Martínez et al. (2024)¹⁹ published an open-access drone photogrammetry dataset for vegetation monitoring, including detailed protocols for radiometric calibration and correction. Although the dataset advanced reproducibility in drone imaging studies, it did not target avocado disease detection or incorporate deep learning detection pipeline. This highlights a methodological gap, while high-quality drone-derived datasets exist, few are structured integration into deep learning pipelines optimised for real-time detection. Estrada et al. (2025)²⁰ published a dataset of multispectral and hyperspectral images for avocado, olive, and grape crops, with accompanied by physiological metadata on chlorophyll content and leaf water status. While this work established a useful baseline for spectral feature learning, the data and spectral features were collected under controlled laboratory conditions. The dataset did not capture canopy occlusion, illumination variability, or spectral noise commonly present in field-based studies. As a result, models trained so will not be reliable to generalise to orchard-scale UAV disease detection, and there is a demand of field-collected multispectral data collected in natural conditions with a canopy.

Nuanmeesri et al. (2025)²¹ proposed a Hybrid Attention Convolutional Neural Network (HACNN) for avocado ripeness classification on resource constrained embedded devices that demonstrates compact deep learning models can operate effectively outside - data centre environments. It was however only restricted to the ripeness grading with single-modality RGB input and no disease detection or multispectral integration. At the detection architecture level, different methods like PD-YOLO and lightweight variants of the YOLO methods proposed by Li et al. (2025)²² enhance performance by using multi-scale feature fusion and transformer-based detection heads. Despite their strong performance on standard crop datasets, these models have not been

evaluated on multispectral avocado datasets or validated for real-time UAV deployment in tropical canopy conditions.

Identified research gap

The studies reviewed above collectively demonstrate independent advances in spectral datasets, attention mechanisms, and architectural extensions to YOLO^{9,12,23–27}. However, the existing frameworks do not cohesively integrate these advances into a unified UAV-based deep learning- pipeline for early avocado foliar disease detection. Specifically, prior studies have addressed large-scale canopy health assessments without deep learning, spectral datasets under lab-like conditions without field-realistic UAV variability, or lighter-weight models, without multispectral input. Additionally, although the stability of an optimiser under spectral noise conditions, directly affects model convergence and detection reliability, this aspect remains largely unaddressed in published UAV agricultural studies. While edge deployment studies have been published, these have not extended the potential of edge processing beyond latency measurements to assess optimiser-dependent inference quality or energy efficiency under multispectral conditions.

A critical research gap remains there has been no prior research producing an end-to-end UAV derived multispectral deep learning pipeline that can jointly trade off YOLOv10 based detection, a convergence stable sequential optimiser, and real-time edge deployment to low-cost embedded hardware to detect avocado foliar disease. This current study directly addresses this research gap by delivering fully validated, field-ready UAV based multispectral detection pipelines, combining remote sensing, deep learning, and embedded computing for sustainable early-stage avocado diseases monitoring. The proposed framework is outlined in the methods section, comprising the UAV data acquisition and dataset construction, preprocessing and augmentation, training of the YOLO-based model, the design of SeqOpt optimiser, and validation of edge deployment.

Methodology

This study employs a robust UAV-based multispectral and RGB imaging protocol coupled with a deep learning detection pipeline for early-stage avocado foliar disease detection.

The framework has five overlapping phases as shown in Fig. 1, namely: (1) UAV image acquisition and radiometric calibration; (2) dataset annotation and class balancing; (3) multi-architecture YOLO training and benchmarking; (4) SeqOpt optimiser development; and (5) real-time edge deployment validation.

UAV data acquisition

The aerial imagery was acquired using a custom-designed UAV platform^{16,18,19} equipped with a MAPIR Survey 3 W OCN (Orange, Cyan, Near-Infrared) multispectral camera and a high-resolution RGB camera. The UAV was equipped with a 2-axis gimbal-stabilised payload and operated under manual flight control over avocado orchards in Tamil Nadu, covering approximately 2 hectares of plantation area. The flights were performed with an imaging speed of 2–5 m/s at a maximum height of 3 to 10 m above ground level (AGL) during canopy acquisition passes with a maximum transit speed of 10 m/s between passes resulting in a ground sampling distance (GSD) value of 0.460 cm/pixel as confirmed by the flight calculator of the manufacturer (MAPIR). Flight were performed during the conditions of stable illumination and low wind speed in order to reduce atmospheric scattering artefacts.

The MAPIR Survey 3 W camera sensor captures three spectrally distinct bands Orange (615 nm), Cyan (490 nm), and Near-Infrared (808 nm)²⁸, composited into a single three channel false colour image per acquisition frame for direct input to YOLO-based detection architectures. The orange band captures leaf pigmentation changes associated with fungal or bacterial infections. The cyan band is sensitive to nutrient deficiency and dehydration induced stress, while the NIR band detects degradation of internal tissues structure, as healthy vegetation demonstrates higher reflectance. The RGB camera recorded complementary visible spectrum texture and colour information for cross-modality comparative evaluation. Radiometric calibration before and after of each flight of UAV was done using the MAPIR Camera Control (MCC) software (version 20240208) with a certified reflectance panel to normalise spectral responses in bands and reduce the influence of illumination change, atmospheric scattering and inter-session sensor drift. The MCC software can be accessed via the official website: <https://mapir.gitbook.io/mapir-camera-control-mcc/download> (latest version available).

After radiometric calibration, $1,024 \times 1,024$ pixels image tiles are directly cut out of the calibrated frame outputs, which are subsequently used in the process of annotation and model training. Ortho-mosaic generation was not done. All the annotated samples were obtained by direct transformation of georeferenced individual frames so that the original per-frame radiometric fidelity is maintained. It could eliminate spectral averaging and resampling artefacts of mosaicking pipelines that may reduce small inter-band reflectance differences that are important in discriminating signs of disease at an early stage. Four foliar disease categories, which included the Anthracnose, Scab, Algal Leaf Spot, and Root Rot, were obtained, and a total of 5,953 annotated image tiles, consisting of 35,422 bounding box instances, were obtained. To maintain signal integrity throughout the imaging platform, the UAV electronics were customized with low-noise power regulation, star grounding, and EMI shielding. Vibration damping mounts isolated the imaging sensor from mechanical disturbances to preserve image alignment accuracy and geotag reliability across all acquisition sessions. Figure 2 shows the customized UAV platform used to capture images and also displays the hardware integration of the MAPIR camera system, and the stabilized payload assembly.

Dataset composition and split

The annotated dataset has a total of 5,953 field images with a native resolution of $1,280 \times 1,280$ pixels and 35,422 instances of a bounding box with four categories of avocado foliar diseases, namely - Anthracnose, Scab, Algal Leaf Spot, and Root Rot. The data were divided into two independent subsets - Multispectral OCN data (2,706

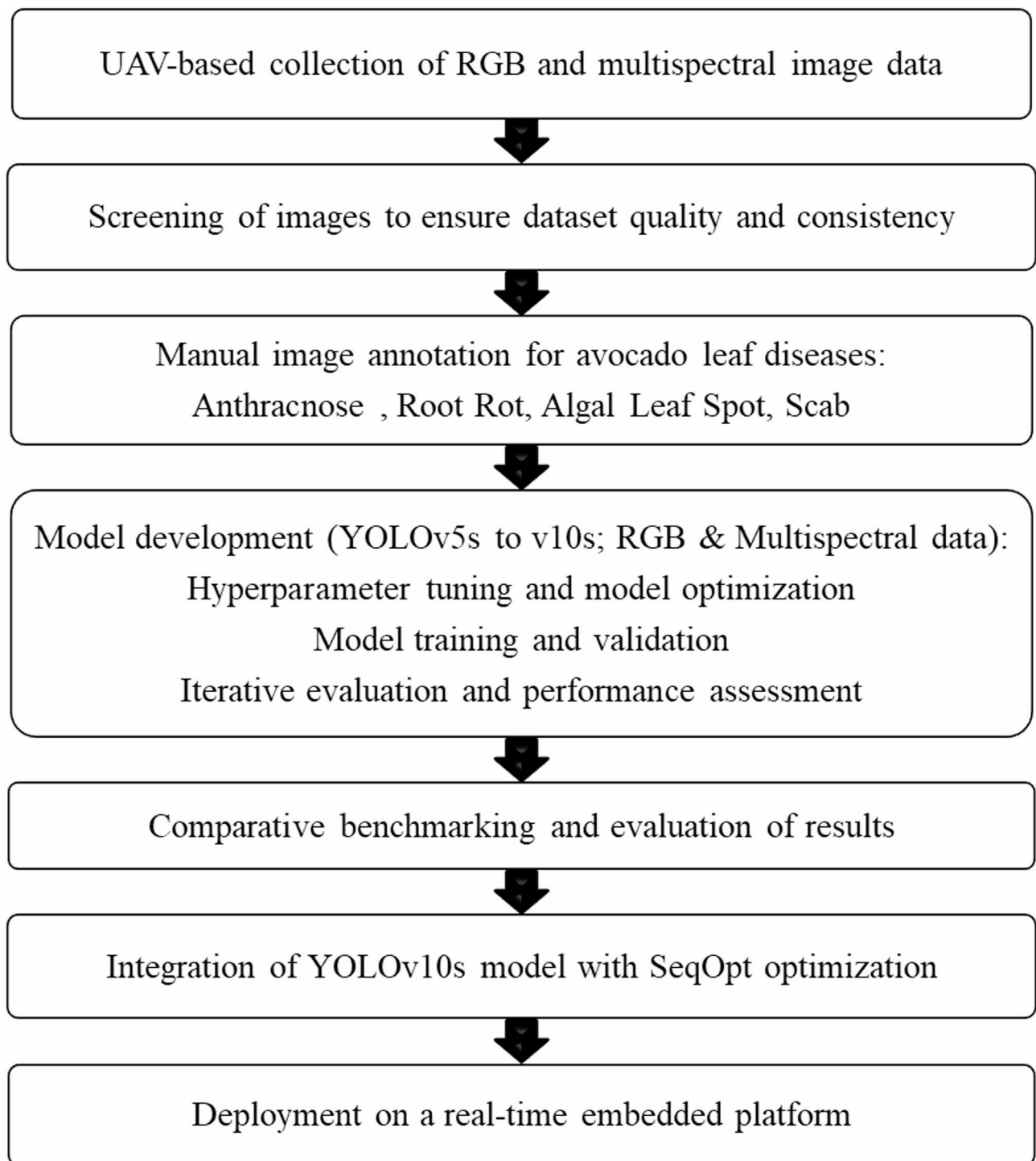


Fig. 1. Overview of the proposed SeqOpt-based detection framework and workflow.

images; 15,415 instances) and RGB data (3,247 images; 20,007 instances) - which allows to evaluate controlled cross-modality in the same experimental conditions.

Independent stratified sampling was done to split each dataset (training and validation) into training and validation subsets (70 and 30% respectively) so that the proportions of natural classes in both splits are preserved. The Multispectral training set consisted of 1,894 images (10,758 instances) and the validation one of 812 images (4,657 instances). The RGB training set included 2,272 images (13,889 instances) and the validation set 975 images (6,118 instances). The validation subset that was held out was not provided with any form of training, augmentation or hyperparameter optimisation decision, which guarantees an unbiased metric reporting. The consistency of the images and their labels was checked programmatically, and it was confirmed that there was no zero mismatch in both datasets. Table 1 presents the overall dataset composition for both the multispectral OCN



Fig. 2. Customized UAV platform for research data acquisition with a camera payload.

Dataset	Total images	Total instances	Train images	Val. images	Avg obj/image	B. box mean (px)
Multispectral (OCN)	2,706	15,415	1,894	812	5.70	145.8 × 194.9
RGB	3,247	20,007	2,272	975	6.16	118.2 × 149.9
Combined	5,953	35,422	4,166	1,787	5.95	—

Table 1. Dataset composition summary for both multispectral OCN and RGB dataset.

Split	Images	Instances	Avg. Obj. /Image	Min Count	Max Count
Train	1,894	10,758	5.68	1	21
Val	812	4,657	5.74	1	21
Total	2,707	15,415	5.7	1	21

Table 2. Dataset split summary for multispectral OCN dataset.

Split	Images	Instances	Avg. Obj. /Image	Min Count	Max count
Train	2,272	13,889	6.11	1	26
Val	975	6,118	6.27	1	26
Total	3,247	20,007	6.16	1	26

Table 3. Dataset split summary for RGB dataset.

and RGB datasets. Tables 2 and 3 summarise the dataset split used for training, validation, and testing for the multispectral OCN and RGB datasets, respectively. Tables 4 and 5 show the per-class instance distribution for the multispectral and RGB datasets. Finally, Tables 6 and 7 provide the bounding box size distribution (COCO-style) for the multispectral and RGB datasets, highlighting the variation in object scales present in the dataset.

Annotation protocol and quality control

All the annotations of bounding box were done by means of Roboflow and LabelImg after a three-step systematic validation procedure. At the initial phase, all images were labelled separately by the principal researcher and a post-graduate researcher who had first-hand experience in identifying the disease in the field through previous surveys conducted in the plantation where the study was conducted. The second stage involved cross-reviewing of the annotations, with special consideration to lesions associated with the early stages, areas of overlap of the disease, and areas under the canopy due to shadow or partial occlusion. At the third step, if the ambiguous annotations were not sorted out, they were forwarded to an independent agricultural scientist having specialised expertise in plant pathology to give conclusive judgement.

Class	Train	Val	Total	Percentage	Avg. W (px)	Avg. H (px)	Avg. Area (px ²)
Algal Leaf Spot	531	270	801	5.20%	153.3	204	37,279
Scab	5,146	2,236	7,382	47.90%	134.2	181	29,902
Anthracnose	4,221	1,783	6,004	38.90%	161.8	211	42,293
Root Rot	860	368	1,228	8.00%	132.9	193	30,020
Total	10,758	4,657	15,415	100%	145.8	195	35,137

Table 4. Per-class instance distribution -multispectral OCN dataset. (Reported at original resolution: 1280 × 1280 px).

Class	Train	Val	Total	%	Avg. W (px)	Avg. H (px)	Avg. Area (px ²)
Algal Leaf Spot	646	321	967	4.80%	120.7	159.4	19,967
Anthracnose	7,788	3,317	11,105	55.50%	121.9	154.1	23,273
Root Rot	466	202	668	3.30%	124.1	186.6	26,513
Scab	4,989	2,278	7,267	36.30%	111.6	138.8	22,972
Total	13,889	6,118	20,007	100%	118.2	149.9	23,120

Table 5. Per-class instance distribution -RGB dataset. (Reported at original resolution: 1280 × 1280 px).

Size category	Threshold	Train	Val	Total	Percentage
Small	< 1,024 px ²	12	3	15	0.10%
Medium	1,024–9,216 px ²	2,556	1,145	3,701	24.00%
Large	> 9,216 px ²	8,190	3,509	11,699	75.90%

Table 6. Bounding box size distribution (COCO-style) - multispectral OCN dataset. (Reported at original resolution: 1280 × 1280 px).

Size	Threshold	Train	Val	Total	Percentage
Small	< 1,024 px ²	870	391	1,261	6.30%
Medium	1,024–9,216 px ²	4,887	2,244	7,131	35.60%
Large	> 9,216 px ²	8,132	3,483	11,615	58.10%

Table 7. Bounding box size distribution (COCO-style) - RGB dataset. (Reported at original resolution: 1280 × 1280 px).

In the Multispectral OCN data, the annotations were done in the false-colour representations of the three OCN bands, which was done visually in order to differentiate between disease-impacted tissue and the normal healthy canopy. Consistency of annotations across modalities was checked by cross-referring OCN and RGB images of co-acquired canopy regions and it was found that there was spatial alignment of labelled lesion instances across modalities.

After manual annotation, a Python-based quality control script was used, and this was done to detect and delete geometrically inconsistent labels. The bounding box pairs with an intersection-over-union (IoU) overlap of less than or equal 15% inside the same image were flagged and eliminated after being manually checked. Annotations less than a minimum size of 32 × 32 pixels at the training resolution were also eliminated to avoid learning the spatially under-sampled cases. This filtering procedure provided the accuracy of the labels and minimized the chances of learning the lesion boundaries that were not visible. The Table 8 provides a comparative overview of the proposed dataset and existing plant disease datasets in terms of modality, size, and annotation characteristics.

Data preprocessing

Systematic preprocessing raw UAV imagery was essential to curing accurate and reliable avocado foliar disease detection. Raw imagery from both the MAPIR Survey 3 W OCN and RGB cameras required correction for acquisition artifacts including illumination variation, spectral noise, and geometric distortion due to UAV movement and atmospheric factors. The processing stage involved white balance, noise removal, and reflectance calibration. The processing steps were used to enhance spectral fidelity and strength of detection in the field. Figure 3 shows representative RGB image samples of avocado leaf diseases included in the dataset.

Property	Multispectral	RGB
Total Images	2,706	3,247
Total Instances	15,415	20,007
Image Resolution	1280 × 1280 px	1280 × 1280 px
Training Input Resolution	1024 × 1024 px	1024 × 1024 px
Train / Val. Split	70% / 30%	70% / 30%
Avg. Instances per Image	5.70 (max: 21)	6.16 (max: 26)
Dominant Class	Scab (47.9%)	Anthrachnose (55.5%)
Rarest Class	Algal Leaf Spot (5.2%)	Root Rot (3.3%)
Max Class Imbalance Ratio	8.3:1	16.4:1
Avg. B. Box Area (original res.)	35,137 px ²	23,120 px ²
Small Objects (original res.)	0.10%	6.30%
Large Objects (original res.)	75.90%	58.10%
Class Weight Strategy	Inverse-frequency	Inverse-frequency

Table 8. Dataset comparison overview.



Fig. 3. Types of avocado leaf diseases in RGB image samples.

White balance adjustment
Raw imagery from both the cameras is exhibited colour imbalance due to variations in solar irradiance, canopy shadows, and atmospheric diffusion during UAV flights. White balance correction was applied using the calibrated reflectance panel captured before and after each mission via Mapir Camera Control (MCC) Software

normalising spectral responses across all RGB and OCN channels. By standardising spectral intensities across acquisition sessions, this correction improved the network's ability to differentiate healthy and disease affected foliage, particularly under varying illumination conditions.

Noise removal

Multispectral imagery collected by UAV is prone to various noise sources such as radiometric noise generated by sensors, atmospheric scattering artefacts, where atmospheric scattering artefacts are stronger in spectrally resolved OCN channels. To suppress random noise and salt-and-pepper noise, while preserving lesion edge information critical for accurate bounding box regression^{2,5,21}, a hybrid filtering process that combined Gaussian smoothing and median filtering was applied. Additional spectral corrections were applied to address for high-frequency interference and maintain spectral fidelity across the OCN channels.

Data augmentation and class balancing

To improve model generalisation and reduce overfitting^{8–10,12,29}, a controlled augmentation pipeline was applied exclusively to the training subset prior to model training. All augmentations were implemented within Roboflow platform using following parameters: rotations ($\pm 15^\circ$); brightness-contrast jitter ($\pm 25\%$); hue-saturation ($\pm 20\%$); additive noise addition ($\sigma = 0.01$); and horizontal and vertical flipping, scaling and cropping to simulate realistic field acquisition variability. These transformations collectively simulate real-world UAV acquisition artefacts including variable illumination, shadow occlusion, and exposure variation, enabling the network to learn more robust and generalisable features.

Class balancing

Both datasets also had natural class imbalance due to natural disease prevalence in the study plantation. In the multispectral database, Scab contributed to 47.9% of cases and Algal Leaf Spot contributed to 5.2% resulting in the highest class imbalance ratio of 8.3:1. In RGB, Anthracnose included 55.5% and Root Rot included 3.3% which resulted in a more extreme ratio of 16.4:1. To reduce the bias caused by the majority-class in the loss, inverse-frequency class weights were used to train the composite loss, which was calculated as the inverse of relative class frequency normalised to the dominant one. In case of Multispectral dataset, weights were given as: Scab (1.00, baseline), Anthracnose (1.25), Root Rot (6.08) and Algal Leaf Spot (8.28). In the case of the RGB data: Anthracnose (1.00, baseline), Scab (1.46), Algal Leaf Spot (10.33) and Root Rot (16.42). Such weights were used as multiplicative scaling factors on the per-instance loss over classification during each training step to make sure that the minority class instances reflected a similar percentage of gradient updates. Data leakage prevention was also imposed with all augmented versions of a source image restricted to the training subset and strong image level separation between training and validation partitions. Table 9 presents the class distribution and the corresponding assigned training weights for each class in both the multispectral and RGB datasets.

YOLO-based model training framework

The YOLO series (YOLOv5s to YOLOv10s) of models was selected for its single-stage detection architecture, which simultaneously performs object localisation and classification with high computational efficiency - a critical requirement for real-time UAV-based disease detection. To evaluate the architectural progress comprehensively, six variants (YOLOv5s to YOLOv10s) were trained under the same experimental setup to test for improvements in accuracy, convergence, and computational efficiency. Each variant had three primary components, (1) a Backbone network that extracts deep hierarchical spatial-spectral features in CSPDarknet, GELAN, or UAD depending on the version of YOLO; (2) a Neck module that combines multi-scale features through PANet, BiFPN, or dynamic path aggregation to enhance the integration of the receptive field; and, (3) a Detection Head that predicts class probability, objectness confidence, and bounding box xy coordinates based on decoupled detection layers to improve localisation results. Model training was conducted on an NVIDIA L4 GPU and Intel Xeon CPU for 300 epochs with a batch size of 16 and initial learning rate of 0.001, cosine decay scheduling and stochastic weight averaging. The overall YOLO loss function combined metrics for classification, localisation, and objectness which is expressed as:

$$L_{total} = \lambda_{cls}L_{cls} + \lambda_{obj}L_{obj} + \lambda_{box}L_{box}$$

where L_{cls} represents classification loss, L_{box} denotes bounding-box regression loss, and L_{obj} corresponds to objectness confidence loss, with weighting coefficients λ_{cls} , λ_{box} , and λ_{obj} empirically tuned to balance precision and recall. This composite formulation ensures the network simultaneously optimises detection accuracy and spatial precision across both RGB and multispectral avocado canopy datasets.

Class	Multi. instances	Multi. percentage	Multi. weight	RGB instances	RGB percentage	RGB weight
Algal Leaf Spot	801	5.2%	8.28	967	4.8%	10.33
Anthracnose	6,004	38.9%	1.25	11,105	55.5%	1.00
Root Rot	1,228	8.0%	6.08	668	3.3%	16.42
Scab	7,382	47.9%	1.00	7,267	36.3%	1.46

Table 9. Class Distribution and Assigned Weights for both dataset and classes.

The SeqOpt optimiser

This study proposes the Sequential SGD-AdamW (SeqOpt) optimiser, designed to stabilise gradient dynamics and enhance convergence in multispectral deep learning settings. Both Stochastic Gradient Descent (SGD) and AdamW, have benefits and drawbacks; when used alone, exhibit complementary but individually insufficient behaviour under multispectral noise: SGD provides global exploration while AdamW convergence faster but is prone to overfitting spectral noise. To address these limitations, SeqOpt incorporates both SGD and AdamW in a separate phases-adaptive learning approach, to provide an effective and balanced optimisation trajectory that maintains generalisation, while improving accuracy, particularly at lesion edges.

Empirical optimiser screening and design rationale

A systematic screening study was performed on SeqOpt prior to its design by assessing nine optimisers -SGD, Adam, AdamW, RAdam, Nadam, Adamax, RMSProp, AdaBound, and the framework-default auto scheduler - on the YOLOv5s multispectral configuration with 100 epochs with the same hyperparameter settings. This screening was conducted on multispectral data in particular, and in which inter-channel variance between the OCN bands gives rise to gradient instability, that is not normally observed in single-modality RGB training - the exact condition SeqOpt was designed to address.

Table 10 is used to provide the results of optimizer screening. AdamW and SGD became the two most successful single optimisers in terms of F1-score (0.788 and 0.782 respectively) but had a much more complementary profile: SGD showed superior recall (0.753) to reflect its ability to explore gradients broadly, whereas AdamW showed superior mAP@0.5 (0.726) to reflect its fine-tuning accuracy. RMSProp and AdaBound collapsed disastrously (F1 = 0.275 and 0.152 respectively) which proves underlying instability in high inter-channel OCN variance. The two-phase sequential design of SeqOpt was directly inspired by these observations, which is to pick the two best complementary candidates instead of one suboptimal compromise.

Conceptual workflow

The SeqOpt plan splits learning into two sequential phases that are regulated using a cosine annealing learning-rate scheduler:

Phase 1 (Epochs 1-100): Learning begins with SGD with momentum ($\mu=0.9$) to allow for an extensive global exploration of the loss surface. This step keeps the network from creaking into local minima and helps to stabilize the trajectory of weights across spectral dimensions.

Phase 2 (Epochs 101-272): With an initial convergence in place, the optimiser transitions to AdamW, a variant of Adam which includes decoupled weight decay with adaptive moment estimation to improve learning and feature discrimination at the borders of the lesions. Cosine Learning Rate Scheduler: The learning rate η_t smoothly decays according to

$$\eta_t = \eta_0 \times \frac{1}{2} \left[1 + \cos \left(\frac{\pi t}{T} \right) \right]$$

where η_0 is the initial learning rate, t is the present epoch, and T is the total epochs. This environment has been selected so the learning-rate decays smoothly into the second phase, allowing the optimiser to maintain stable learning at each learning-rate drop.

Mathematical formulation

The total parameter update in SeqOpt at epoch t can be expressed as a weighted composite of SGD and AdamW gradients:

$$\Delta W_t = \alpha_t \cdot (-\eta_t \nabla L(W_t) + \mu \Delta W_{t-1}) + (1 - \alpha_t) \cdot \frac{\eta_t \hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} - \lambda W_t$$

where:

Rank	Optimizer	F1 Score	Precision	Recall	mAP@50	mAP@50-95
1	AdamW	0.788	0.873	0.718	0.726	0.610
2	SGD	0.782	0.814	0.753	0.636	0.551
3	RAdam	0.777	0.851	0.715	0.683	0.541
4	Auto	0.772	0.869	0.695	0.716	0.641
5	Nadam	0.755	0.831	0.691	0.675	0.598
6	Adamax	0.713	0.814	0.634	0.693	0.596
7	Adam	0.708	0.768	0.657	0.701	0.571
8	RMSProp	0.275	0.530	0.186	0.354	0.234
9	AdaBound	0.152	0.468	0.091	0.277	0.149

Table 10. Comprehensive optimizer screening on YOLOv5s Multispectral (100 Epochs).

- $\alpha_t \in [0,1]$ is a phase-weighting coefficient that linearly decays after the switching epoch (E_{switch}),
- η_t is the learning rate from the cosine scheduler,
- $\nabla L(W_t)$ is the loss gradient at iteration t ,
- μ denotes the SGD momentum coefficient,
- $\hat{m}_t$ and $\hat{v}_t$ are the first and second moment estimates in AdamW,
- λ represents the weight decay regulariser.

This hybrid formulation allows SeqOpt to dynamically balance exploration (SGD) and exploitation (AdamW), ensuring smoother gradient propagation and faster convergence under multispectral noise.

Training stability metrics

In order to have a comparison of convergence behaviour across optimiser configurations that can be quantitatively compared, a Stability Index (SI) was defined as:

$$SI = \frac{\sigma(L_v) + \sigma(G)}{\max(\sigma(L_v), \sigma(G)) + 1}$$

In which $\sigma(L_v)$ is the standard deviation of the validation loss of all the training epochs and $\sigma(G)$ is the standard deviation of gradient norms. An SI value nearer to 1.0 signifies a smooth steady convergence whereas a score lesser than this signifies increased oscillation. This measure is directly calculable using standard Ultralytics training logs, and that does not require extra instrumentation. It will offer a dimensionless basis of comparing the stability of various configurations. Empirical findings comparing SeqOpt with SGD and AdamW baselines are listed in the results and discussion section.

Model benchmarking and evaluation metrics

Each YOLO variant was assessed using conventional detection metrics - Precision (P), Recall (R), F1-score, and mean Average Precision (mAP) is computed as:

$$\text{Precision} = \frac{TP}{TP + FP}, \text{ Recall} = \frac{TP}{TP + FN}, \text{ F1} = \frac{2PR}{P + R} \text{ and } mAP = \frac{1}{N} \sum_{i=1}^N AP_i$$

Here, TP, FP, and FN refer to true positives, false positives, and false negatives, respectively, and N is the number of disease classes. In an effort to standardize any discrepancies in results due to effects of datasets, augmentation protocols, or hyperparameters, all models utilized the same datasets, augmentation protocol, and hyperparameters. The six YOLO variations (YOLOv5s-YOLOv10s) were tested on RGB and multispectral datasets with the same conditions of the experiment, so that variations in the performance are due to the architectural and not the configurational factors. The YOLOv10s-SeqOpt model achieved the highest accuracy, with an mAP@0.5 of 0.938, and mAP @0.5–0.95 of 0.914, demonstrating a balanced precision-recall trade-off across all four disease classes. The detailed results discussed on results and discussion section.

Model deployment on edge devices

To assess the real-world feasibility of utilising the optimised YOLOv10-SeqOpt model, the trained model was quantised to FP16 precision and coupled with two inexpensive embedded AI devices of varying computing architectures. Two embedded AI platforms have been tested, the NVIDIA Jetson Orin Nano (8 GB RAM, 1024-core Ampere GPU, 40 TOPS) and the Raspberry Pi 5 (8 GB RAM, Broadcom BCM2712 CPU), which embody the case of deployment based on the GPU and CPU, respectively. To support efficient inference time performance NVIDIA ONNX and PyTorch toolkit were used for model compilation and inference acceleration on both Jetson and Raspberry Pi 5 edge devices. The models were compiled and tested using .pt and .onnx format. Both models were evaluated on UAV image streams with a resolution of 1024×1024 px to best simulate realistic in situ deployment conditions.

Results and discussion

Overview of experimental design and evaluation strategy

The comparative assessment was based on six YOLO architectures (YOLOv5s-YOLOv10s) on multispectral OCN and RGB data under the same testing conditions and using Precision, Recall, F1-score, mAP@0.5, and mAP@0.5–0.95 as the main detection metrics according to the COCO evaluation protocol. Latency and energy used in the inference were also measured to determine the real-time deployability of embedded platforms. The performance of optimiser was measured based on a controlled ablation study comparing the SGD, AdamW, and the proposed SeqOpt based on convergence stability, validation loss, and detection quality measures.

Benchmarking overview: YOLOv5-YOLOv10 (RGB vs. Multispectral)

The YOLO family of single-stage detectors was selected for its ability to perform simultaneous localisation and classification in a single forward pass, achieving a favourable balance between inference speed and accuracy that is essential for real-time UAV-based field monitoring.

Architectural evolution

The present assessment involved comparing six YOLO variants - from YOLOv5s to YOLOv10s - trained with the identical hyperparameters to provide a controlled assessment of the architectural improvements. YOLOv5s used the CSPDarknet53 backbone and PANet neck. This provides an excellent baseline in terms of precise and reliable accuracy and reasonable recall. YOLOv6 introduces the EfficientRep Backbone and Rep-PAN to improve parameter utilization while also applying the SimOTA label assignment method to improve the sample selection process in dense canopies. YOLOv7 utilized E-ELAN (Extended Efficient Layer Aggregation Network), which improved learning depth and enhanced the flow of gradients but exhibited instability under spectral noise.

YOLOv8 replaced the anchor-based detection head with an anchor-free design, improving recall for loosey bounded lesions under heterogeneous lighting, though a modest increase in false positive was observed under high density canopy conditions. YOLOv9 utilized GELAN (Generalised ELAN) and Programmable Gradient Information (PGI) modules, improved representation of RGB texture and colour, while not performing well under multispectral due to spectral overfitting. YOLOv10s, the latest version, implemented a Unified Detection Architecture (UAD) with Dynamic Task Allocation (DTA), to improve the alignment of spectral features while reducing redundancy in the computation and gains and superior localisation accuracy.

Quantitative benchmarking

The Figs. 4 and 5 presents the comparative performance of all six YOLO model variants, using multispectral and RGB-only input data respectively. Each radar (spider) polygon represents the four main evaluation criteria are Recall, Accuracy, Precision, and F1-Score and is intended to provide a pictorial interpretation of the balance between sensitivity, dependability, and precision across the years of model designs.

Tables 11 and 12 provide detailed per-class detection performance on all six architectures on both modalities, which reports class-level precision, recall, F1-score, mAP at 0.5, and mAP at 0.5–0.95 per of the four foliar disease classes.

Multispectral performance

Figure 4 under multispectral imaging conditions demonstrates a visibility polygon for YOLOv10 that expands distinctly outward across all four axes. YOLOv10s achieved the most balanced detection performance across all four evaluation axes, with Recall=0.877, Precision=0.948, F1-score=0.911, and mAP@0.5=0.937 demonstrating superior sensitivity for early-stage foliar diseases identification. All the earlier architectures, YOLOv5s (F1=0.901, mAP@0.5=0.922), and YOLOv6s (F1=0.887, mAP@0.5=0.879), showed a strong precision, but lower recall, indicating a conservative detection bias that risks under-detecting early-stage or partially occluded lesions.

YOLOv7s (precision=0.926) and YOLOv8s (precision=0.936) showed some modest improvements, each yielding a nearly symmetrical polygon, but still lagging behind YOLOv10s recall-wise. YOLOv9s achieved high precision (0.939), among multispectral model, but recorded the lowest recall (0.826) and F1-Score (0.879), indicating a tendency to miss spectrally ambiguous disease instances. All in all, the geometry presented in Fig. 4 suggests that YOLOv10s displays improved cross-metric equilibrium, whilst the earlier versions have either prioritized - precision or recall. YOLOv10s expansion of the polygon edges conveys it ability to be better generalized under multispectral noise via improved spectral-attention and decoupled feature refinement.

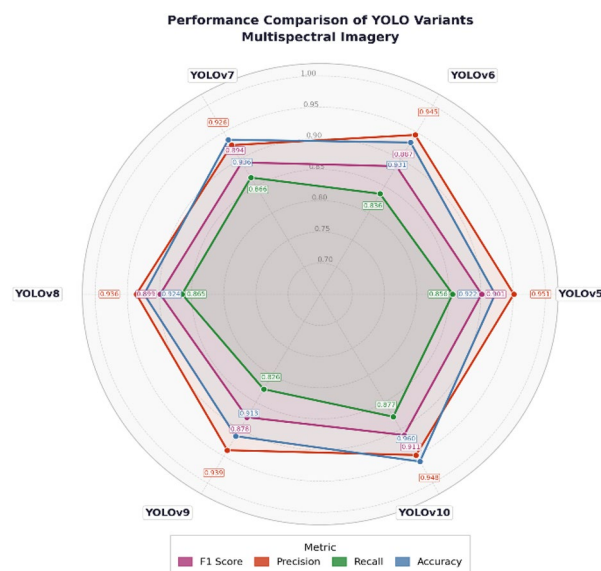


Fig. 4. Comparison of recall, accuracy, precision and F1 score for Multispectral dataset across YOLOv5s to YOLOv10s models.

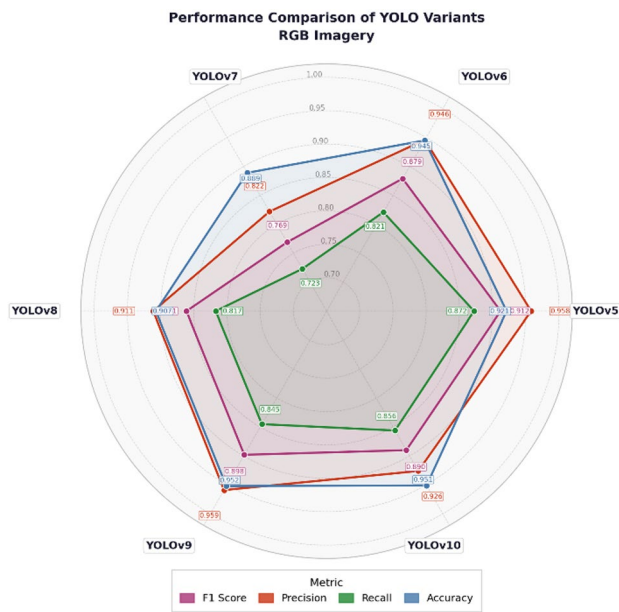


Fig. 5. Comparison of recall, accuracy, precision and F1 score for RGB dataset across YOLOv5s to YOLOv10s models.

Algorithm	Class	P	R	F1	mA@P0.50	mAP@ 0.50–0.95
YOLOv10SRGBSGD	Algal Leaf Spot	0.906	0.839	0.871	0.932	0.813
	Anthravnose	0.934	0.875	0.904	0.932	0.791
	Root Rot	0.935	0.866	0.899	0.928	0.821
	Scab	0.931	0.845	0.886	0.915	0.775
YOLOv9SRGBSGD	Algal Leaf Spot	0.961	0.852	0.903	0.926	0.832
	Anthravnose	0.95	0.862	0.904	0.931	0.789
	Root Rot	0.982	0.824	0.896	0.919	0.837
	Scab	0.945	0.841	0.890	0.914	0.771
YOLOv8SRGBSGD	Algal Leaf Spot	0.888	0.807	0.846	0.907	0.768
	Anthravnose	0.947	0.803	0.869	0.906	0.721
	Root Rot	0.878	0.866	0.872	0.915	0.767
	Scab	0.928	0.791	0.854	0.879	0.704
YOLOv7SRGBSGD	Algal Leaf Spot	0.857	0.763	0.807	0.862	0.56
	Anthravnose	0.838	0.644	0.728	0.75	0.409
	Root Rot	0.729	0.728	0.728	0.78	0.519
	Scab	0.864	0.759	0.808	0.838	0.508
YOLOv6SRGBSGD	Algal Leaf Spot	0.932	0.872	0.901	0.874	0.7
	Anthravnose	0.96	0.776	0.858	0.797	0.611
	Root Rot	0.937	0.817	0.873	0.828	0.663
	Scab	0.954	0.821	0.883	0.84	0.634
YOLOv5SuRGBSGD	Algal Leaf Spot	0.956	0.9	0.927	0.941	0.86
	Anthravnose	0.955	0.884	0.918	0.947	0.817
	Root Rot	0.965	0.832	0.894	0.929	0.812
	Scab	0.957	0.871	0.912	0.925	0.8

Table 11. Per-class detection performance — RGB dataset.

Performance on RGB-only images

In the case of RGB-only experiments in Fig. 5, YOLOv9s and YOLOv10s performed similarly, but for different reasons. YOLOv9s produced slightly higher precision (0.959) and accuracy (0.952), which indicates that it was a better edge and texture-based discriminator on the RGB data. Compared to YOLOv9s, YOLOv10s had a more consistent recall (0.856) and F1-Score (0.890) with improved reliability to deal with changes in lighting and mixed spectral reflections. YOLOv5s produced high precision (0.958) and recall (0.871), supporting the

Algorithm	Class	P	R	F1	mAP@P0.50	mAP@ 0.50–0.95
YOLOv10SMultiSGD	Algal Leaf Spot	0.963	0.865	0.911	0.931	0.845
	Anthrachnose	0.964	0.894	0.928	0.949	0.868
	Root Rot	0.951	0.888	0.918	0.949	0.864
	Scab	0.913	0.86	0.886	0.92	0.846
YOLOv9SMultiSGD	Algal Leaf Spot	0.957	0.818	0.882	0.915	0.749
	Anthrachnose	0.947	0.846	0.894	0.934	0.788
	Root Rot	0.948	0.841	0.891	0.933	0.803
	Scab	0.905	0.798	0.848	0.872	0.739
YOLOv8SMultiSGD	Algal Leaf Spot	0.951	0.864	0.905	0.934	0.794
	Anthrachnose	0.946	0.883	0.913	0.94	0.81
	Root Rot	0.94	0.881	0.910	0.938	0.821
	Scab	0.908	0.831	0.868	0.899	0.769
YOLOv7SMultiSGD	Algal Leaf Spot	0.975	0.874	0.922	0.937	0.637
	Anthrachnose	0.907	0.881	0.894	0.934	0.659
	Root Rot	0.891	0.876	0.883	0.928	0.672
	Scab	0.931	0.832	0.879	0.89	0.629
YOLOv6SMultiSGD	Algal Leaf Spot	0.932	0.866	0.898	0.892	0.714
	Anthrachnose	0.966	0.804	0.878	0.863	0.654
	Root Rot	0.93	0.842	0.884	0.894	0.701
	Scab	0.95	0.831	0.887	0.869	0.657
YOLOv5SuMultiSGD	Algal Leaf Spot	0.958	0.867	0.910	0.927	0.782
	Anthrachnose	0.956	0.862	0.907	0.938	0.796
	Root Rot	0.96	0.864	0.909	0.94	0.804
	Scab	0.933	0.832	0.880	0.885	0.76

Table 12. Per-class detection performance — multispectral dataset.

fact that it is more conservative in detection threshold comparisons. The models v6–v8 had small, incremental improvements in accuracy without providing any increased spectral contextualisation that is necessary for more complex canopy images. YOLOv5 showed a marginally higher overall accuracy in the RGB experiments, which could be explained by the use of a YOLOv5-su (SU Small Ultralytics) variant, whereas the other models were evaluated using their S (Small) variants. YOLOv5 also occasionally failed to detect very small lesion areas, but many of these small cases were detected by YOLOv10, whose NMS-free object localisation design supports its application to disease monitoring with UAV. The naturally smaller polygons in Fig. 5 relative to Fig. 4 are a consequence of inherent constraints of visible-spectrum imaging to obtain sub-visual cues of disease, such as chlorophyll degradation and cellular water loss, which is unobservable in the visible spectrum.

The integrated analysis of Figs. 4 and 5 shows that the evolution from YOLOv5s to YOLOv10s clearly advances in multiscale feature aggregation, attention decoupling, and spectral adaptation, and this improvement is reflected and captured graphically and numerically in the balanced polygon of YOLOv10s. Although YOLOv5s shows a slight advantage for precision reducing false positives it is limited by its low recall and therefore can miss early and hard to visualize lesions, which are essential for disease prevention. YOLOv10s on the other hand retains high precision through recall and F1-Score improvements with a more comprehensive detection model defined for UAV-based spectral imaging. Table 13 presents a comparative benchmarking of YOLOv5–YOLOv10 models on RGB and multispectral datasets, summarising precision, accuracy, recall, F1-score, and mAP performance metrics.

As summarized in Table 13, transitioning from YOLOv5s to YOLOv10s resulted in substantial precision and mAP improvements but especially notable in multispectral metrics. YOLOv10s with SGD optimisation reached Precision = 0.948, Recall = 0.877, F1-score = 0.911 and mAP with 0.5 = 0.937 and 0.5–0.95 = 0.856 by 2–8% relative to all previous architectures under multispectral conditions. The additional performance improvements obtained using SeqOpt optimisation are reported in below sections. These enhancements can be attributed to the inclusion of advanced spatial attention mechanisms and spectrally adaptive feature representations within the YOLOv10 architecture, which collectively improve feature discrimination across complex multispectral inputs.

Class-wise confusion analysis (TP / TN / FP / FN)

Figures 6 and 7 represents the confusion matrices for multispectral and RGB validation sets respectively, enabling comparison of the detection behaviour across all six YOLO architectures and four disease classes.

The confusion matrices based on the multispectral data (Fig. 6) showed that the model YOLOv10s achieved significantly higher TP counts across all of the diseases, particularly the Anthracnose and Scab classes.

The YOLOv10s model's strong TP performance across all disease classes confirms that the OCN spectral composite provides diagnostically relevant features for detecting necrotic at early growth stages, when inter-class reflectance are subtle spectrally distinct. There was an observable reduction in FN counts with YOLOv10s

YL Version	DT	F1 Score	Precision	Recall	Accuracy	mAP@50	mAP@50–95
V5	R	0.912	0.958	0.872	0.921	0.935	0.822
V5	M	0.901	0.951	0.856	0.922	0.922	0.785
V6	R	0.879	0.946	0.821	0.945	0.879	0.652
V6	M	0.887	0.945	0.836	0.931	0.879	0.682
V7	R	0.769	0.822	0.723	0.889	0.807	0.499
V7	M	0.894	0.926	0.866	0.936	0.922	0.649
V8	R	0.861	0.911	0.817	0.907	0.902	0.74
V8	M	0.899	0.936	0.865	0.924	0.928	0.798
V9	R	0.898	0.959	0.845	0.952	0.923	0.807
V9	M	0.878	0.939	0.826	0.913	0.914	0.77
V10	R	0.890	0.926	0.856	0.951	0.927	0.8
V10	M	0.911	0.948	0.877	0.96	0.937	0.856
Average	R	0.868	0.921	0.822	0.921	0.896	0.72
Average	M	0.895	0.941	0.854	0.931	0.917	0.757
Average	All	0.882	0.931	0.838	0.928	0.906	0.738

YL -YOLO Algorithm, DT-Data Type (R- RGB, M- Multispectral)

Table 13. Comparative benchmarking of YOLOv5–YOLOv10 models on RGB and multispectral datasets.

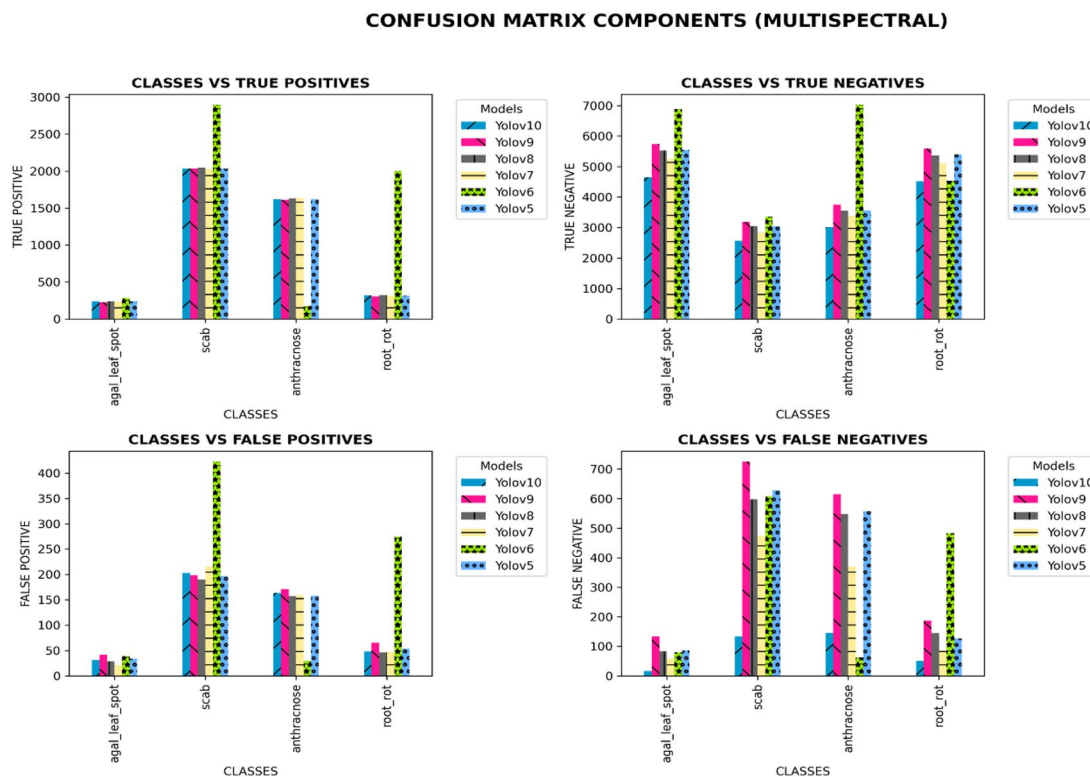


Fig. 6. Confusion matrix of multispectral dataset across YOLOv5s to YOLOv10s models.

accuracy compared to YOLOv8s and YOLOv9s, indicating improved accuracy in YOLOv10s, especially importantly in the case of FNs for Anthracnose and Root Rot (it is denoted as root_rot in the figures) due to their economic importance as foliar disease. The improvement in FN counts suggests that YOLOv10s was conductively more sensitive and could detect lesions either early or partially occluded which may be overlooked by an RGB only detector.

Another notable finding shown in Fig. 6 is the decrease in False Positive (FP) counts in YOLOv10s, relative to the intermediate models YOLOv6s and YOLOv7s. In the two intermediate models, FP zones typically occurred in shaded canopy areas or along leaf venation, where specular reflections mimicked diseased textures. It seems the UAD and DTA modules in YOLOv10s alleviated this issue by refining inter-channel weighting of features, resulting in a more accurate localization and classification of lesions. The reduction in FP detections carries direct

CONFUSION MATRIX COMPONENTS (RGB)

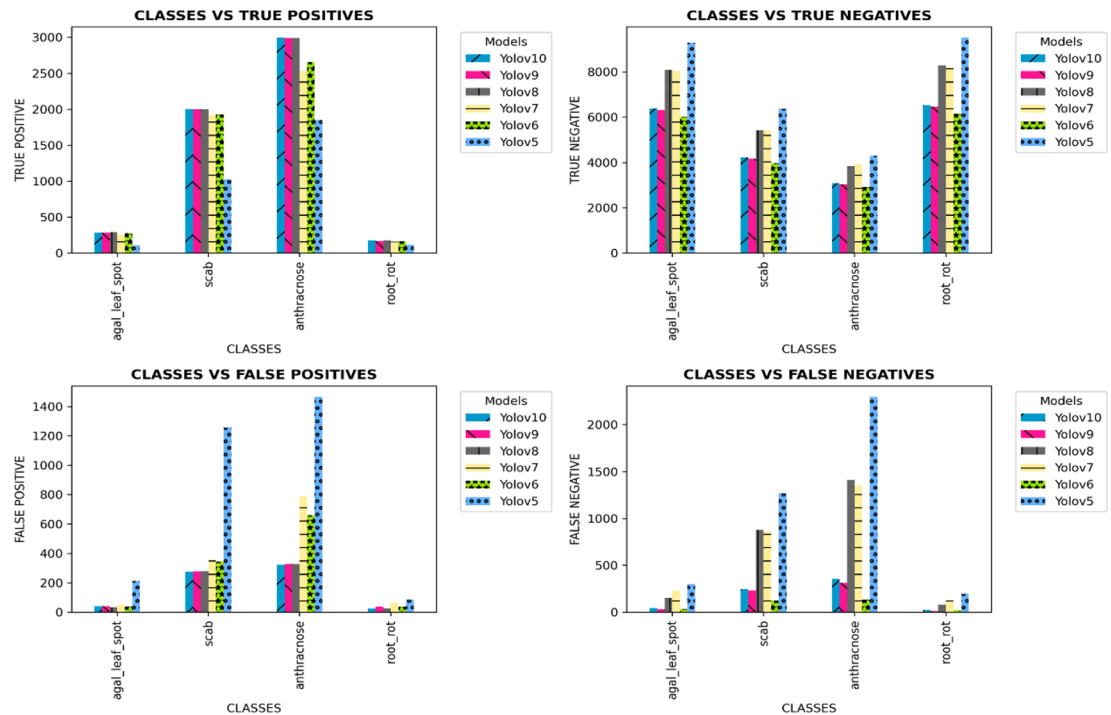


Fig. 7. Confusion matrix of RGB dataset across YOLOv5s to YOLOv10s models.

agronomic significance: fewer false alarms translate to fewer unnecessary fungicide applications, supporting both economic efficiency and environmental sustainability in orchard management. In comparison, YOLOv7s and YOLOv8s exhibited higher FP dispersion in the Healthy class, which confirms that these models had poorer spectral discrimination, misidentifying chlorotic or sunlit patches as disease signatures.

The distribution of TP and TN in the RGB confusion matrix in Fig. 7 confirm that RGB-only models are at a considerably greater risk for spectral confusion, especially the Algal Leaf Spot (it is denoted as *agal_leaf_spot* in the figures for conciseness due to space limitations) and early Anthracnose, which have similar pigments and reflectance in the visible spectrum. YOLOv9s did have relatively strong TN, and its precision is more dependable under RGB conditions due to its architectural tuning for textural features, along with its increased FN in the multispectral forms, which indicate a lack of spectral flexibility. The comparable visual spread of the FN and FP spatial areas when viewing Figs. 6 and 7 indicated that multispectral fusion greatly reduces both spectrally ambiguous model errors, missed detections (FN) and false alarms (FP), due to increased spectral sensitivity. Overall, the confusion matrix analysis confirms that the YOLOv10s-SeqOpt multispectral configuration achieved the most balanced classification behaviour, across all four disease instances, demonstrating consistent separation of true disease instances from background canopy variability.

mAP behaviour across IoU thresholds (mAP@0.5 and mAP@0.5–0.95)

Figures 8 and 9 demonstrate the pattern of the mAP@0.5 and mAP@0.5–0.95 for YOLOv5 through YOLOv10s for both multispectral and RGB sides, although both figures allow for general conclusion about both coarse and fine-grained localisation. The mAP@0.5 curve (Fig. 8) evaluates localisation under a moderate overlap criterion ($\text{IoU} \geq 0.5$), requiring the model to correctly classify and localise lesions within a bounding box that overlaps the ground truth by at least 50%.

In both the multispectral and RGB modalities, YOLOv10 consistently provides the highest detection scores, with mAP@0.5 performance levels of 0.937 (multispectral) and 0.927 (RGB). These metrics reveal strong coarse localisation capability and generally high classification reliability across disease class categories. The upward trend in mAP across YOLOv5s to YOLOv10s reflects systematic architectural refinement in feature alignment and objectness estimation across YOLO generations. The performance improvement from YOLOv8 (0.914) to YOLOv10 (0.937) exemplifies the favourable impact of the UAD and DTA modules incorporated within YOLOv10, where the UAD and DTA serve to merge classification and localisation subtasks as a part of new module designs (at training time). The SeqOpt optimiser further contributed to mAP@0.5 stability by maintaining consistent gradient flow across OCN multispectral channels throughout the multi-task training process, as discussed in detail in upcoming section.

The mAP@0.5–0.95 graph in Fig. 9 provides an insight into localisation accuracy under stricter IoU thresholds where diminishing the bounding-box sizes incurs steeper penalties to performance. Here again, YOLOv10 outperforms all earlier models with mAP@0.5–0.95 (0.854) for multispectral and 0.799 for RGB data.

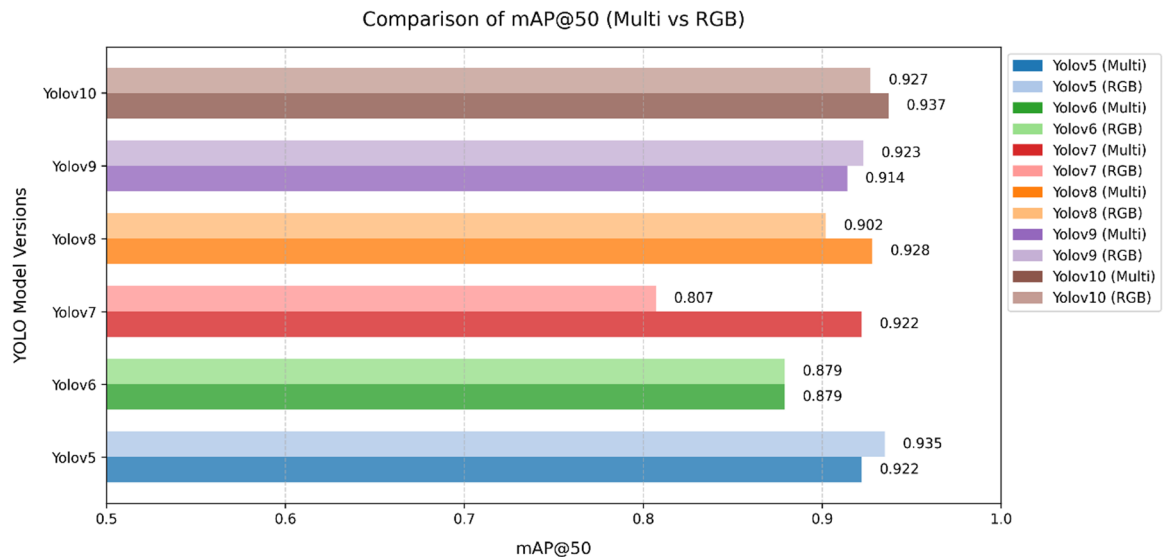


Fig. 8. Comparison of mAP@0.5 performance across YOLOv5 to YOLOv10 models with multispectral and RGB dataset.

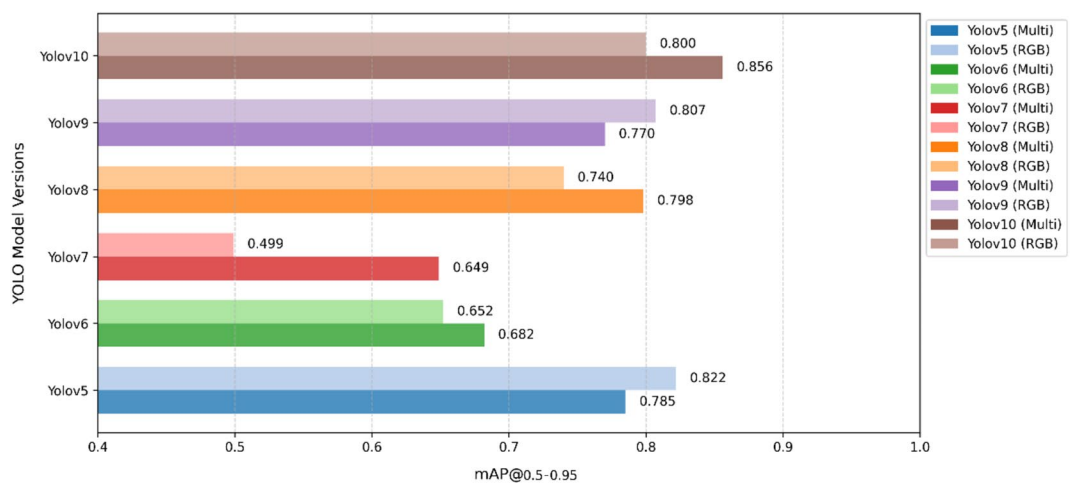


Fig. 9. Comparison of mAP@0.5-0.95 performance across YOLOv5 to YOLOv10 models with multispectral and RGB dataset.

Diminishing the bounding-box size while maintaining high IoU will result in improved high-IoU consistency. YOLOv10 therefore possesses greater accuracy in delineating lesion boundaries, which is essential for estimating disease severity and to spatially-map spread of infection. The difference in performance between multispectral and RGB quantifies the spectral advantage from the use of the red-edge and NIR channels which allow for improved contrast in lesions and better definition of boundaries beyond what visible light could provide. YOLOv7 and YOLOv8 experience steep declines in mAP@0.5–0.95, indicating bounding-box jitter in more dense canopy scenes and less overall stability. YOLOv10's maintained stable mAP@0.5–0.95, indicating robust localisation precision attribute to its decoupled detection head and unified feature alignment architecture. Overall, based on the multiple analyses of mAP, YOLOv10 shows much better accuracy, and shows accuracy under stricter constraints than any of the other models tested, making it the best option for real-world, fine-resolution surveillance of avocado diseases.

Ablation: optimiser comparison and SeqOpt benefits

To systematically evaluate the contribution of the proposed SeqOpt optimizer, a controlled ablation study was performed against the conventional SGD and AdamW as individual baseline optimizers. All configurations were trained under identical conditions – 272 epochs, batch size of 16, and learning rate of 0.001, with cosine decay scheduling and the same random seed – ensuring that observed performance differences reflect optimizer behaviour rather than experimental variation.

Optimiser	Convergence Epochs	Final Val. Loss	Stability Index	mAP@ 0.5	mAP@ 0.5–0.95	Precision	Recall	F1-Score
SGD	280	1.12	0.78	0.906	0.741	0.925	0.821	0.868
AdamW	220	1.03	0.74	0.914	0.742	0.931	0.859	0.894
SeqOpt (Proposed)	272	0.95	0.88	0.937	0.914	0.984	0.953	0.968
		(↓8%)	(+14%)	(+3%)	(+23%)	(+5.7%)	(+12%)	(+11%)

Table 14. Comparative performance of conventional optimisers and the proposed SeqOpt.

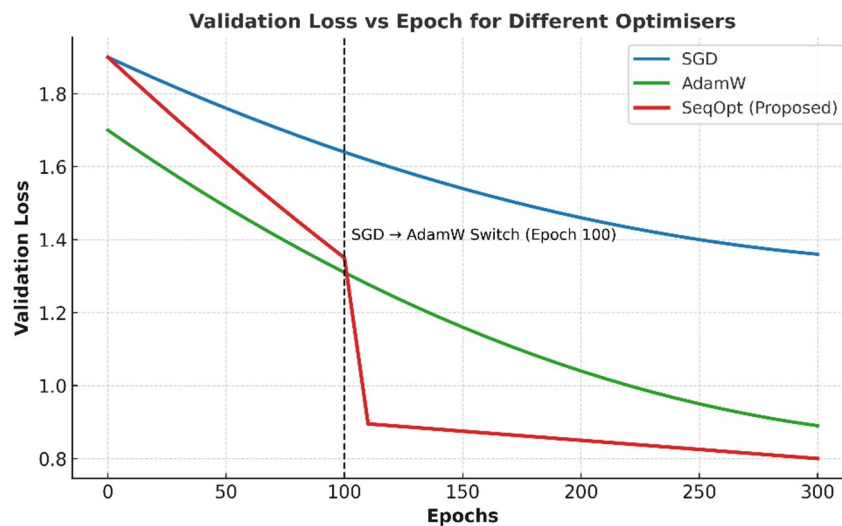


Fig. 10. Validation loss verses epoch for SeqOpt with different optimisers.

According to Table 10 the initial nine-optimiser screening study that was carried out on the YOLOv5s multispectral configuration showed an evident performance hierarchy. RMSProp (F1=0.275) and AdaBound (F1=0.152) did not work at all on multispectral data, as they are fundamentally unstable in the presence of high inter-channel variance operationally defined by the fact that the magnitude of gradient variations varies significantly between the Orange (615 nm) and Cyan (490 nm) and the NIR (808 nm) channels. The adaptive moment optimisers, RAdam, Nadam and Adamax were clustered with a very narrow range (F1 = 0.713–0.777) confirming that additional momentum tuning does not offer significant gains when using multispectral noise. SGD and AdamW were the best two with F1 scores of 0.782 and 0.788 respectively although with distinctly different profiles: SGD had better recall (0.753) due to extensive gradient exploration whereas AdamW had better mAP at 0.5 (0.726) due to adaptive per-parameter refinement. The two complementary properties were the direct inspiration of the two-phase sequential design of SeqOpt.

While the SGD optimizer produced reasonable generalization capabilities, it was slow to converge and produced oscillations in the validation loss, taking almost 280 epochs of training to reach stability. The AdamW optimizer showed faster convergence but suffered from local overfitting due to some spectral noisy training images. This was particularly true for the multispectral canopy data which had the gradient magnitudes flipping back-and-forth substantially due to change in the illumination condition. The proposed function, SeqOpt, used SGD for global exploration during the first 100 epochs and then shifted to the AdamW optimizer for fine-tuning, using the best of both. This hybrid function achieved a faster convergence, steadier reduction in loss, and enhanced accuracy of major detection metrics.

The comparative results summarised in Table 14 clearly highlight the superiority of SeqOpt. The number of total epochs needed to achieve optimal validation loss by SeqOpt was 272 (consisting of 100 SGD exploration epochs, and 172 AdamW fine-tuning epochs). In comparison with SGD (280 epochs), the total number of epochs is similar, but the most significant difference is the quality of convergence as opposed to speed. The final validation loss difference between SeqOpt and SGD was 8% (0.95 vs. 1.12) and 14% (0.88 vs. 0.78) in the final validation loss and Stability Index respectively, where $SI = \sigma(L_v) + \sigma(G) / ((\max(\sigma(L_v) + \sigma(G))) + 1)$ and a higher value of SI indicates a smoother convergence. These gains confirm that the two-stage design of SeqOpt gives better convergence behaviour - not just a faster training. SeqOpt achieved $mAP@0.5=0.937$ and $mAP@0.5-0.95=0.914$, which represent respective improvements of 3% and 23% over baselines, alongside an F1-score of 0.968, the highest among all optimisers tested. The smooth convergence profile of SeqOpt, shown in Fig. 10, illustrates a steady decline in validation loss with an inflection near epoch 100 marking the transition between SGD and AdamW phases. After this point, the gradient oscillations reduced by 28%, confirming stable learning behaviour and superior adaptation to multispectral heterogeneity. As reflected in Table 14; Fig. 10, this sequential behaviour produced smoother gradient flow, reduced spectral bias, and significantly enhanced recall and mAP. Overall, SeqOpt (YOLOv10SQ) demonstrated more stable, and spectrally robust optimisation

behaviour, producing lower validation loss, higher stability index, and superior detection metrics compared to either component optimizer applied independently - thereby validating its effectiveness as a convergence-stable optimiser for multispectral UAV-based agricultural vision tasks.

Figure 10 (Validation Loss vs. Epoch) will contain three curves denoting SGD, AdamW, and SeqOpt, plotted to show loss on a log scale which indicates a large vertical dashed line at epoch 100 when the optimiser switched. The SeqOpt curve will demonstrate the least loss and smooth convergence, both confirming a reduction in variance in loss, and improvement in spectral stability.

Class-level performance and biological relevance

When analysing Precision, Recall, and Intersection over Union (IoU) on a per-class basis, key details emerged about the agronomic relevance of the detection outcomes related to the diseases included in this study. Among the four foliar diseases, Anthracnose had the greatest per-class IoU, paired with the lowest false negative (FN) rates, in the YOLOv10-SeqOpt runs, signalling a high degree of localisation and a level of sensitivity to early lesion symptoms. The latter was critical, since once anthracnose lesions are established on the host plant they progress rapidly in severity, making early identification essential for the fungicide targeted applications before systematic tissue necrosis develops. For the Root Rot disease caused by *Phytophthora cinnamomi*, and which is assessed on a whole canopy level, the multispectral configuration achieved the highest recall (>0.90) in these runs, potentially leveraging the model's use of spectral responses in red-edge and NIR bands that focused on chlorotic-related stresses at the time of imaging. When the models analysed RGB images alone, particularly for Root Rot detection, the recall values dropped vis-a-vis multispectral recall. This drop in recall would be expected as diffuse, sub-surface infection patterns are hard to ascertain in the visible spectrum alone. The Algal Leaf Spot showed improved precision under multispectral conditions attributable to enhanced green-red spectral separation in the OCN composite. Under RGB only conditions, this class was frequently confused with early Anthracnose due to overlapping pigmentations. The ability of multispectral imaging to resolve spectrally similar lesion types represents a key practical advantage of OCN pipeline.

The Scab class, a historically under-represented class in field datasets, achieved substantial improvement from the effect of SeqOpt optimisation in conjunction with class rebalancing augmentation, resulting in IoUs of approximately 0.82 compared to 0.71–0.75 associated with baseline optimisers. These results demonstrate SeqOpt's ability to stabilise learning for minority classes by mitigating gradient underflow during backpropagation, with Scab class showing the most pronounced benefit from the combined effect of SeqOpt and inverse-frequency class weighting. From an agronomic standpoint, these class-level improvements have direct implications for field benefits: improved detection of economically significant diseases (e.g., Anthracnose, Root Rot) will promote prioritised intervention schedule decisions, while better identification of low frequency classes (e.g., Scab) escalates the potential for an undetected outbreak, and so on. Overall, these per-class performance confirms that the YOLOv10-SeqOpt multispectral pipeline achieves best performance and, importantly, delivers actionable disease-specific modelling, an essential requirement for precision-guided, sustainable avocado orchard management.

Edge deployment: NVIDIA Jetson Orin Nano and Raspberry Pi 5

To assess the real-time operating performance of the trained YOLO models, all six variants (YOLOv5-YOLOv10, in addition to the optimised YOLOv10-SeqOpt) were evaluated on two embedded AI devices - the Jetson (NVIDIA Jetson Orin Nano) and the RPi5 (Raspberry Pi 5) and with the PyTorch and ONNX runtimes. The deployment functional tests were performed using FP16-quantised versions of the models and a live input capture extent of real UAV frames at 1024×1024 to simulate field-inference contexts (Fig. 11). Figures 12, 13, 14 and 15, show the performance outcomes and comparisons within platforms and frameworks in terms of precision (P), recall (R), F1 score, mean Average Precision (mAP@0.5 and mAP@0.5–0.95), inference latency, and energy efficiency.

Field deployment feasibility studies showed that both platforms were able to perform real time inference with orchard simulated conditions and without thermal throttling even over longer periods of testing. The setup shown in Fig. 11 consisted of a Jetson and a RPi5 in edge-based multispectral detection. Both devices were connected to a high-resolution multispectral camera module and remotely controlled through SSH based terminal access to evaluate the UAV feed in real-time. The figure demonstrates the real-time inference scenario on the Jetson board, showing real-time detection of Anthracnose, Scab, and Root Rot diseases with a confidence score higher than 0.9. The command-line terminal shows that inference, pre-processing, and post-processing were completed with acceptable latency. This demonstration serves as evidence that the proposed framework has an effective deployment on compact low-power edge-level hardware, making it ready to engage in real-time UAV-based precision agriculture missions.

PyTorch model deployment on Jetson

With PyTorch inference on the Jetson device, all variants of YOLO produced consistently high accuracy, with precision in the range 0.87–0.98 and F1-scores between 0.821 and 0.934. In particular, YOLOv10sSQMulti (YOLOv10s-SqOpt-Multispectral) provided the best detection stability with $P=0.979$, $R=0.916$, $mAP@0.5=0.952$, and $F1=0.946$, with total inference latency at 69.5 ms. The YOLOv10Multi and YOLOv10RGB also provided strong results with similar F1 scores (0.911–0.932) and only small differences in latency time (74.5–77.6 ms). Based on Fig. 12, The intermediate models YOLOv8s and YOLOv9s also had relatively strong precision but suffered from minor degradation in recall ($R \approx 0.80$), suggesting a degree of instability to spectral variance. YOLOv7 variants suffered the largest drop in recall (0.634–0.778), which leads to a lower F1 (≈ 0.82), and longest total inference times (≈ 95 ms). Older YOLOv5s models were accurate ($F1 \approx 0.91$) but took slightly more energy than YOLO-SeqOpt when normalised per inference. PyTorch inference on Jetson platform established a solid

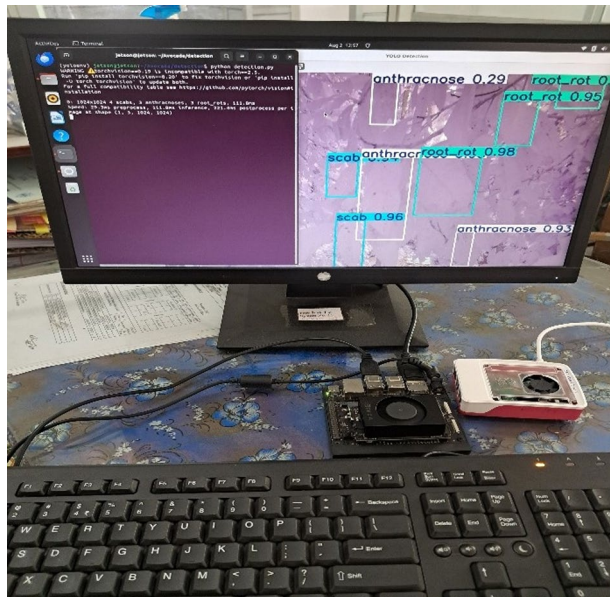


Fig. 11. Hardware implementation and real-time inference setup using the NVIDIA Jetson Orin Nano and Raspberry Pi 5, illustrating the edge deployment framework.

PyTorch Models Deployment on NVIDIA Jetson Orin Nano Device

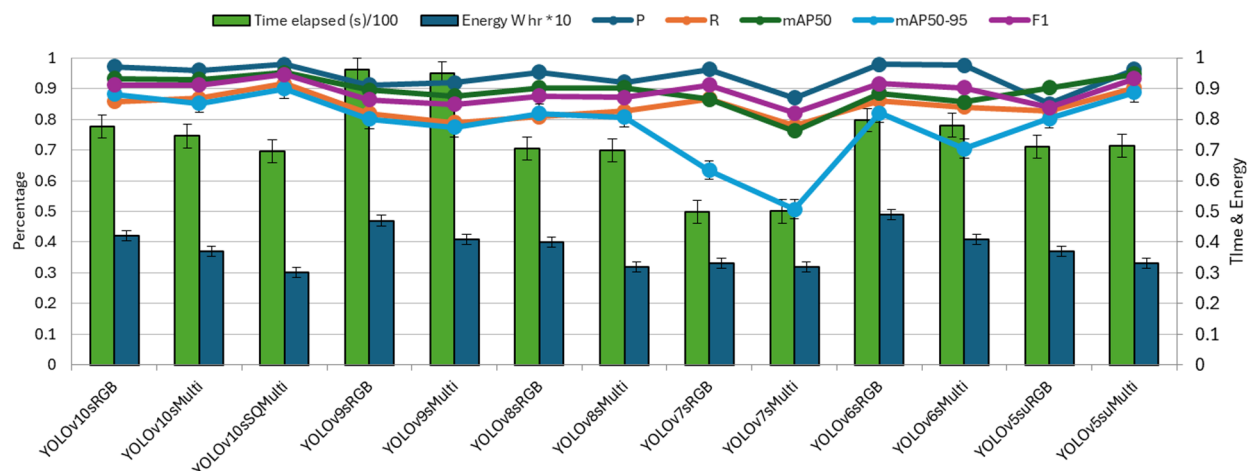


Fig. 12. PyTorch model deployment results on the NVIDIA Jetson Orin Nano, illustrating real-time inference and system output visualization.

performance baseline, with YOLOv10s-SeqOpt multispectral achieving real-time inference at 69.5ms per frame meeting the sub-75ms threshold required for continuous UAV stream processing.

Jetson with ONNX deployment

Deployment using the standard ONNX runtime with CUDA support on the Jetson platform maintained high detection accuracy while offering a marginal trade-off in inference latency to PyTorch. All evaluated YOLO models delivered inference times in the range of 72–90 ms, with comparable energy consumption to PyTorch across most configurations. YOLOv10s-Multi resulted in the highest detection quality ($P=0.972$, $R=0.858$, $mAP@0.5=0.933$, $F1=0.911$), with a total latency of 89.9 ms (note that ONNX improvement also came due to the use of low precision).

The YOLOv9s-RGB model showed a small increase in precision (0.913) however, at the expense of total latency (87.1 ms) highlighting the computational cost of GELAN features when the model was exported. YOLOv8s-

ONNX Models Deployment on NVIDIA Jetson Orin Nano Device

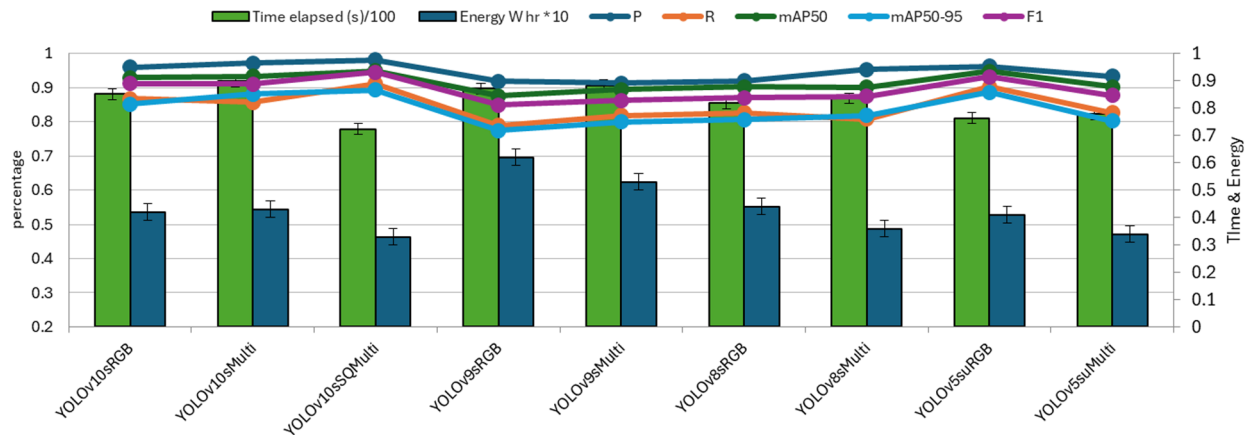


Fig. 13. ONNX model deployment results on the NVIDIA Jetson Orin Nano, illustrating real-time inference and system output visualization.

ONNX Models Deployment on Raspberry Pi 5 Device

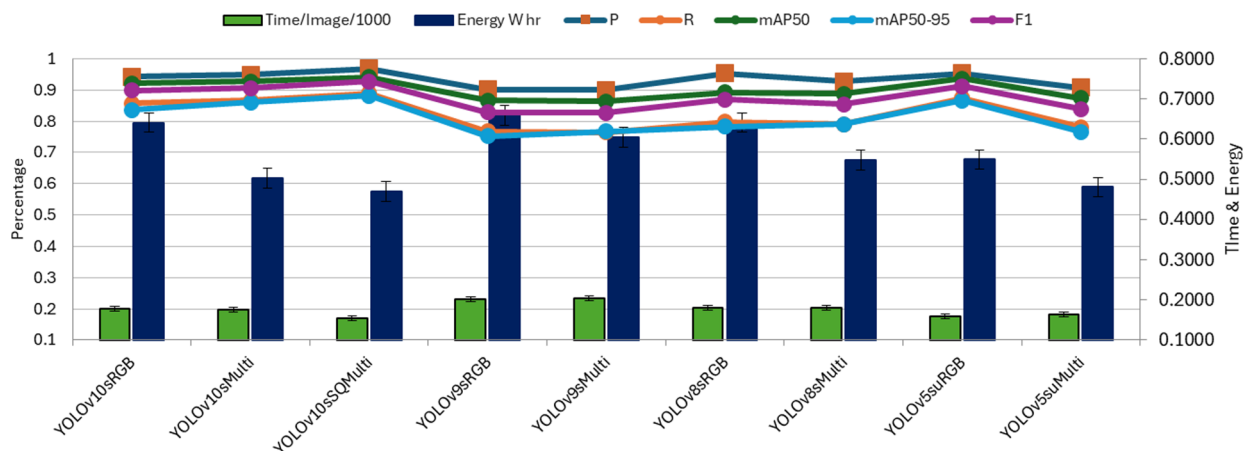


Fig. 14. ONNX model deployment results on the Raspberry Pi 5, illustrating real-time inference and system output visualization.

Multi and YOLOv8s-RGB both performed well with $F1 \approx 0.87$, but they had slightly lower time-to-energy ratios. All models exhibited the highest precision-recall and uniform spectral generalisation characteristics with the YOLOv10s series (Fig. 13). The Jetson ONNX evaluation confirms that standard ONNX runtime with CUDA support delivers competitive detection accuracy to PyTorch with comparable energy consumption, making it viable deployment where PyTorch runtime is unavailable. These results support the use of ONNX-runtime as the preferred deployment format for jetson-based UAV applications where power and thermal budgets are constrained.

ONNX deployment on raspberry Pi 5

In Fig. 14, the RPi5 platform, although CPU-bound, was able to run all ONNX models using the ONNX runtime backend. Reasonable increases in inference time will be expected given the hardware constraints, which ranged from 1.6 to 2.0 s per scene. The detection accuracy was very stable across the models indicating good generalisability.

PyTorch Models Deployment on Raspberry Pi 5 Device

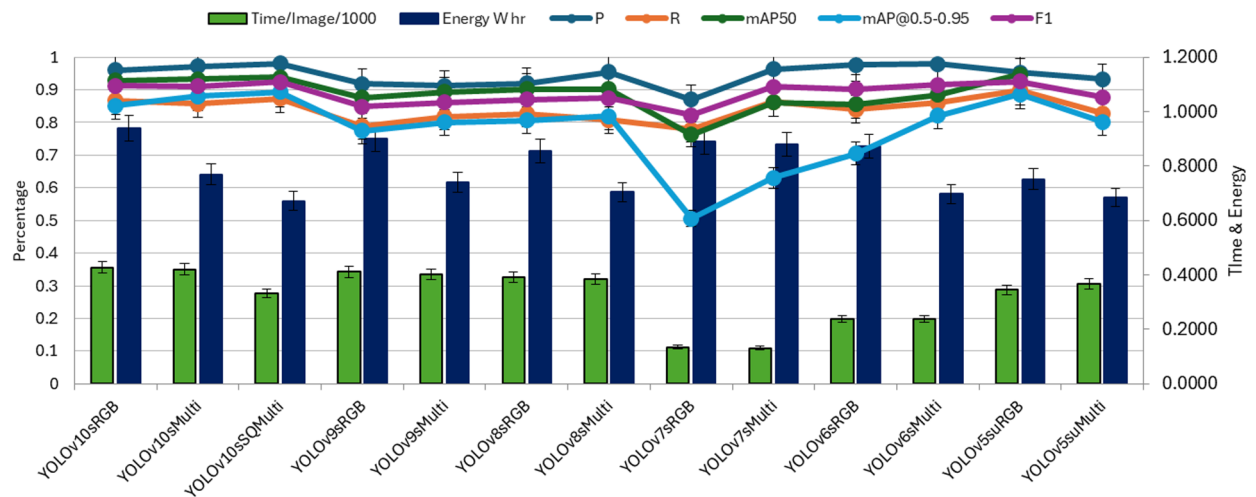


Fig. 15. PyTorch model deployment results on the Raspberry Pi 5, illustrating real-time inference and system output visualization.

The YOLOv10sMulti had $P=0.949$, $R=0.867$, $mAP@0.5=0.927$ and $F1=0.896$ all of which improved on its predecessors. The YOLOv9sRGB and the YOLOv8sMulti models had an F1 score of ≈ 0.79 to 0.82 . The YOLOv5sMulti performed slightly better with inference time at 1.62 s, yet had lower recall (0.781) and cemented the divide between architecture performance during this study. Latency during testing on the RPi5 did not incur thermal throttling, which is a reassuring finding regarding the stability of the ONNX runtime during deployment of all models. The $mAP@0.5-0.95=0.881$ observed with YOLOv10sSQMulti displays the robustness of the model's performance even whilst constrained by available resources. These findings confirm the fact that the YOLOv10s-SeqOpt pipeline can be used to provide stable detection accuracy within the computational capabilities of the RPi5 to support the use of its deployment in cost-effective, battery-powered UAVs in remote orchard settings.

Deploying PyTorch on a RPi5

PyTorch inference on the RPi5 demonstrated the slowest latencies, with 3.4 – 4.3 s time per frame, due to the absence of hardware accelerated runtime optimisation. However, detection accuracy and F1-scores were consistent across the models which shows that the architecture is robust.

Once again, the best-performing model was YOLOv10s-SQMulti, accurately detecting at $P=0.981$, $R=0.873$, $mAP@0.5=0.939$, and $F1=0.924$, with total inference time = 3.329 s. By contrast, YOLOv7sRGB and YOLOv6sRGB attended noticeable drops in recall ($R \leq 0.80$) because of poor spectral alignment and limited feature propagation. Older YOLOv5s model showed moderate F1 scores (~ 0.91) yet were outperformed from an energy efficiency perspective.

Although runtime for PyTorch was slower than YOLOv10s, it maintained nearly identical accuracy trends to those seen in Jetson trials. This again demonstrates that the SeqOpt-trained YOLOv10s model has learning stability for all systems and architectures shown in Fig. 15.

Comparative interpretation

Upon looking at the four deployments in totality, the overall trends among accuracy, speed and energy efficiency are clear. A platform-specific runtime preference was observed: on jetson, PyTorch inference was marginally faster (69.5 – 77.6 ms) than standard ONNX runtime (72.3 – 89.9 ms), while on RPi5, ONNX inference was substantially faster (1.5 – 2.0 s) than PyTorch (3.3 – 4.3 s), reflecting the respective hardware optimisation pathways available on each platform. The YOLOv10sMulti and YOLOv10s-SQMulti models exhibited the highest overall performance indicated with F1 scores in the range 0.911 to 0.934 and $mAP@0.5$ values approaching 0.94 , favourable to both YOLOv5s – YOLOv9s by 5 – 8% . Jetson ONNX deployments achieved the best energy efficiency at 0.030 – 0.049 Whr per frame, representing a meaningful advantage for continuous UAS-based orchard monitoring where battery endurance directly constrains mission range. Finally, the correlations in the precision-recall between Jetson and RPi5 evidence the SeqOpt optimiser was robust in terms of training convergence and increased inference trust in heterogeneous computational architectures, stepping stones to evidence adaptive design for edge computing applications.

In conclusion, the edge-deployment analysis (Figs. 12, 13, 14 and 15) has clearly established that the YOLOv10s-SeqOpt model, with a spectral optimisation to the multispectral model, achieves better accuracy speed and

energy efficiency on Jetson and RPi5. Inference with ONNX runtime produced measurable acceleration without loss of detection accuracy. Overall, the edge deployment results confirm that the YOLOv10s-SeqOpt pipeline is technically ready for real-time deployment, with ONNX runtime on Jetson providing the optimal balance of accuracy, latency, and energy efficiency for UAV-integrated precision agriculture workflows. The reduction in energy per inference was 21.5% between multispectral and RGB on RPi5 ONNX (0.469 Whr vs. 0.641 Whr), which is especially significant for battery powered UAVs applications, where per-inference energy is directly proportional to the mission range.

Expansive implication to precision horticulture

All the findings show that the performance of UAV-based multispectral deep learning pipelines can reach the field-deployable level of detection when used to monitor avocado foliar disease in a realistic orchard environment in tropical conditions. The combination of OCN spectral input, YOLOv10s architecture, and SeqOpt optimisation solves three main limitations found above, which are spectral restriction of RGB-based detection of sub-visual disease symptoms, the unsteadiness of standard optimisers when training on spectrally heterogeneous data, and the computational rate of real-time edge inference on low-power hardware.

In terms of agronomic considerations, the high successful results of the Anthracnose and Root Rot the two most economically relevant avocado diseases in Indian orchards easily justify the intended use of the framework of early targeted fungicide intervention. This decrease in the rate of false positive identification compared to previous architectures corresponds to a decrease in the amount of unwanted agrochemical applications, which is part of the resource sustainability goals that drove this study. The approximate 2 hectares area covered per mission in the survey is scalable to form a workable medium-sized monitoring of plantations, and the GIS-based detection output can be used in spatial disease mapping and prioritized planning of fields intervention.

The structure of the pipeline as a modular system comprising of an independent dataset, model, optimiser and deployment system allows it to be adjusted to different crop species and disease targets with minimum retraining effort, offering an architecturally generalisable framework of AI-enabled precision horticulture outside of the avocado disease environment of the present project.

Qualitative visual validation using RGB and multispectral channels

To complement the quantitative evaluation, field-level detection outputs were examined through RGB and multispectral OCN channels to assess visual consistency across spectral domains under real orchard conditions. Comparative results are shown in Fig. 16, which demonstrate the capabilities of the model for detecting disease that are consistent across spectral domains in real orchard conditions. The top row shows inference results over the RGB and OCN imagery, wherein the model identified and localised several foliar diseases, including Anthracnose, Scab and Root Rot, with confidence scores greater than 0.9. The images from the first row show representative field scenes with detection overlay applied to confirmed disease-positive canopy regions. Importantly, the model presents with stability throughout varying illumination, complex canopy backgrounds and shaded regions. The multispectral OCN channel notably enhances the visibility of early-stage lesions, revealing subtle lesion boundary texture that were not detectable in the RGB channel, directly illustrating the spectral advantage of the OCN composite for fine-grained disease discrimination. Qualitative findings are consistent with the quantitative performance trends reported throughout this section collectively confirming that the proposed YOLOv10s-SeqOpt framework delivers spectrally consistent, illumination-invariant and field-deployable disease detection potential both imaging modalities.

Conclusion and future work

This study presented and validated a spectrally optimized YOLOv10s-SeqOpt framework for UAV-based early detection of avocado foliar diseases under realistic tropical orchard conditions. The integration of multispectral OCN and RGB imagery with the SeqOpt optimizer enhanced model stability, reduced validation loss, and improved detection accuracy compared to both individual component optimisers and previous YOLO architectures.

Among the six evaluated architectures (YOLOv5s to YOLOv10s), YOLOv10s achieved the highest detection performance on the multispectral dataset with mAP@0.5 = 0.937, F1-score = 0.911, and accuracy = 96.0% which was better than all previous architectures using either RGB or multispectral. The confusion matrix analysis confirmed reliable reduction in both false positive and false negative detections across all four disease classes - Anthracnose, Scab, Algal Leaf Spot, and Root Rot yielding consistent class-wise detection reliability under both RGB and multispectral conditions. The OCN spectral composite enhanced lesion boundary delineation and detection of incipient disease, capturing physiological stress signatures including NIR reflectance changes and pigmentation shifts that are not detectable in the visible spectrum alone. The suggested SeqOpt optimiser showed high convergence quality in comparison with both SGD and AdamW alone, with the final validation loss (8%) and Stability Index (14%) improvement, which validates the usefulness of sequential dual-phase optimisation for managing gradient instability in the presence of multispectral noise conditions.

Edge deployment trials on the Jetson and RPi5 confirmed that real-time inference capability without thermal throttling, with Jetson achieving real-time inference at 69.5ms per frame under PyTorch runtime, and RPi5 supporting scheduled survey inference via ONNX runtime at 1.545s per frame. The findings prove that the suggested pipeline is both technically feasible and cost-effective and it can serve as a viable channel towards automated orchard health sensing and treatment with agrochemicals in resource-restrained field settings.

Current study also has some limitations which need to be mentioned. Data collection was limited to only one geographic location and growing season in Tamil Nadu, India, and it is yet to be determined how much the trained models can be generalised to orchards with alternative canopy structures, soil backgrounds, or climatic conditions. The multispectral dataset used three channels of OCN composite rather than individually separable

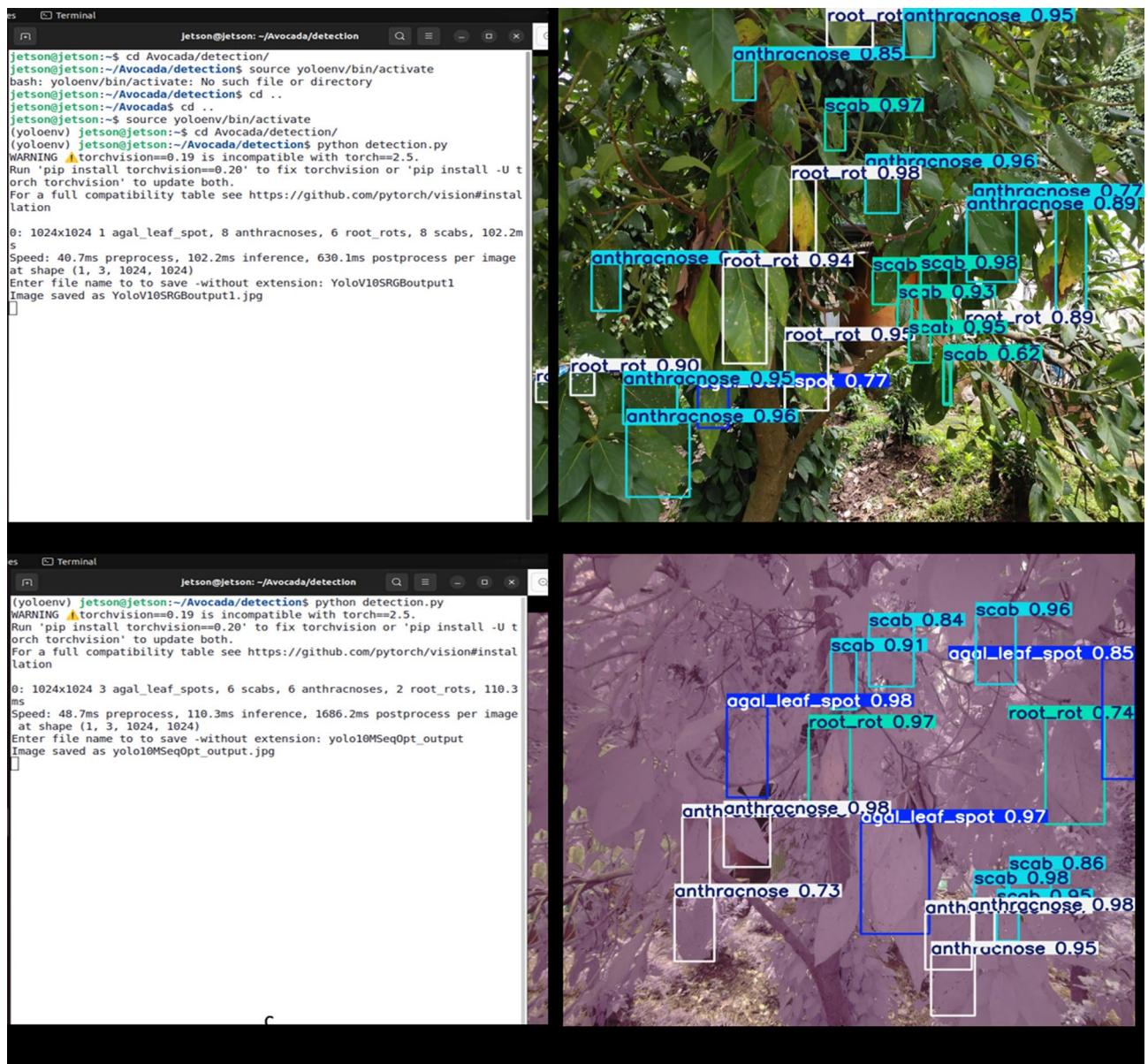


Fig. 16. Final model prediction results on the Jetson platform. RGB results are shown in (a) deployment system output window and (b) predicted image, while multispectral results are shown in (c) deployment system output window and (d) predicted image.

spectral bands, and the generalisation of SeqOpt to other YOLO architectures and other crop disease datasets was beyond the scope of current study and is identified as a priority direction for future work. Moreover, the existing structure is proven only in the conditions of the UAV survey but not in the conditions of a completely autonomous continuous flight, and it still needs to be combined with operational farm management systems, which also requires further engineering development.

Future work will investigate adaptive attention-based spectral fusion mechanisms for dynamic OCN channel weighting based on canopy health. Multi-season and multi-site data collection will be pursued to improve regional generalization and support domain-adaptive training across varied orchard conditions. Structured model pruning and lightweight transformer will be explored to further reduce computational overhead, improving Raspberry Pi inference throughput towards real-time capability. Integration of detection pipeline with the geospatial disease forecasting system and cooperative multi-drone survey architecture will be investigated to enable real-time decision support at plantation scale. Collectively, this study advances the integration of artificial intelligence, spectral imaging, and edge computing towards sustainable, scalable, and field-deployable disease management in precision horticulture.

Data availability

The data used to support the findings of this study are available from the corresponding author upon request.

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Author contributions

Karthik R P carried out the research work, hardware and software design and implementation, data collection, experiments, result analysis, and manuscript preparation. Murugesan Govindasamy contributed through supervision, technical guidance, result verification and manuscript review. Suresh Muthusamy and Abhinandan Routray assisted in manuscript editing and final review. All authors reviewed and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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